

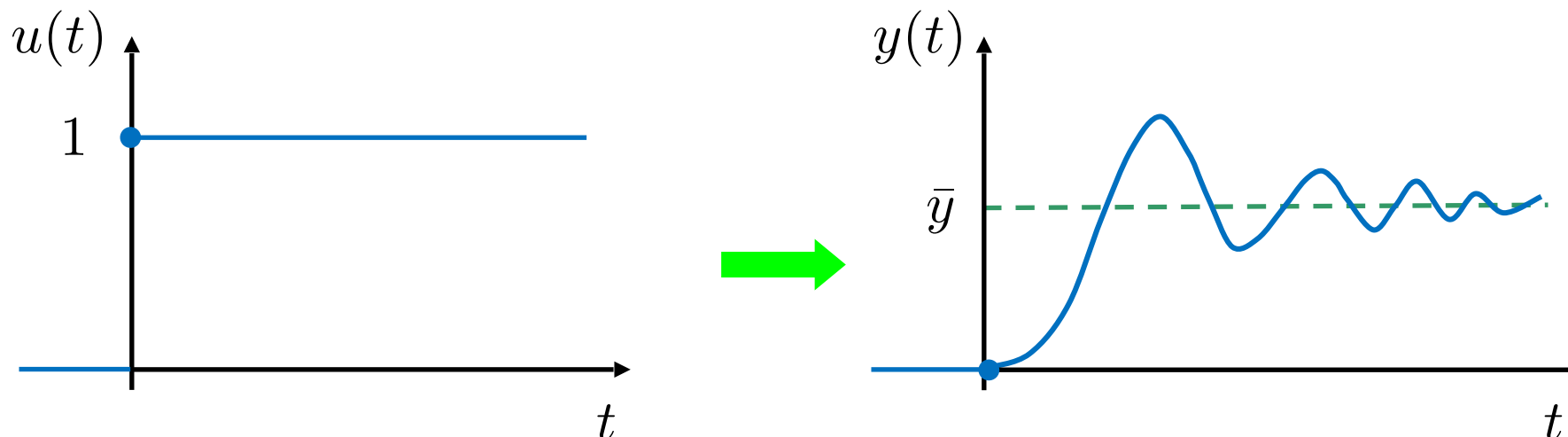
**034IN - FONDAMENTI DI AUTOMATICA -
FUNDAMENTALS OF AUTOMATIC
CONTROL
A.Y. 2024-2025
Part VII: Step-Response Analysis**

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Step Response

$$x(0) = 0; \quad u(t) = 1(t)$$

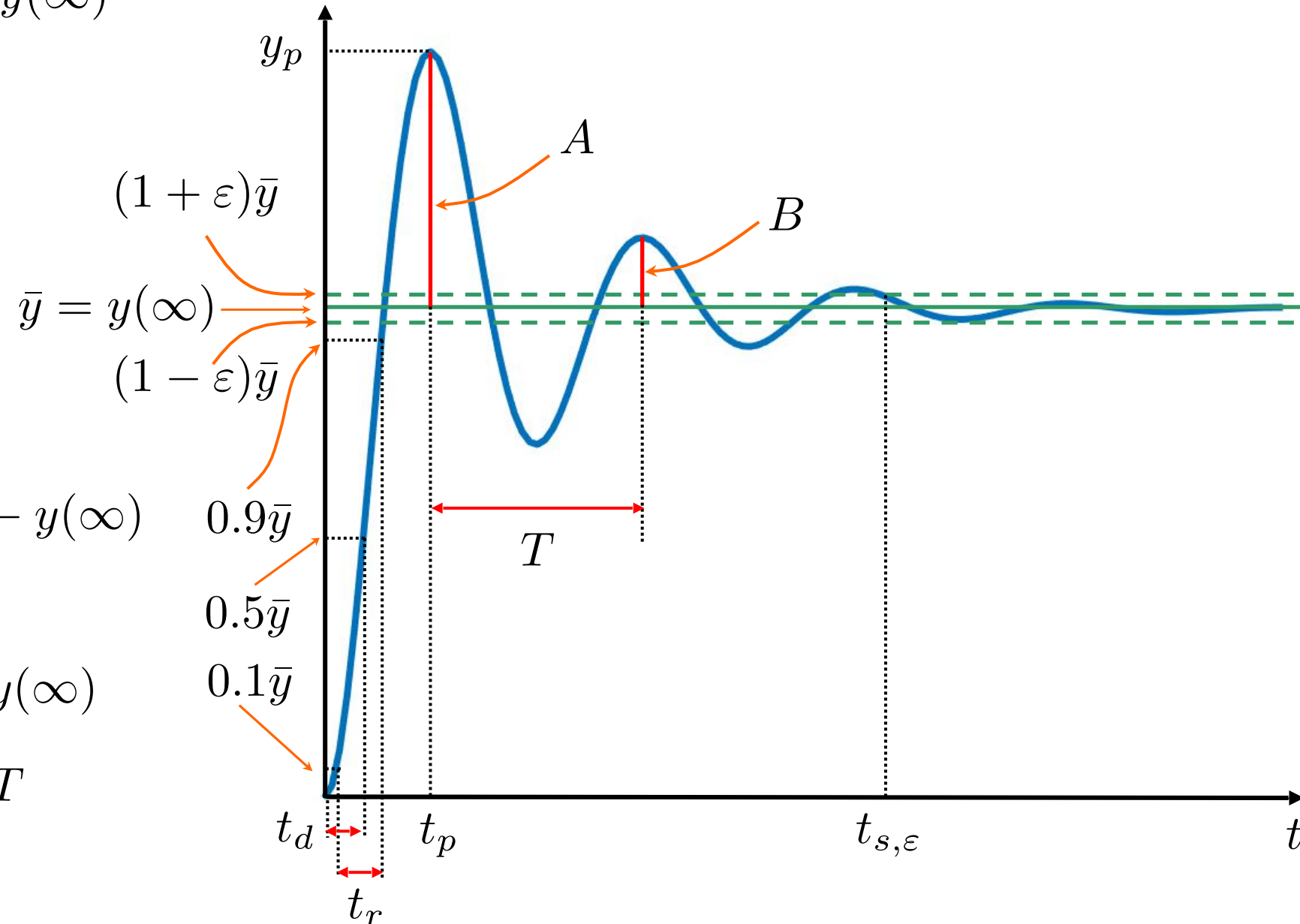


- ◆ For asymptotically stable systems, the step response describes the way the systems "moves" from an equilibrium to another
- ◆ The characteristics of the step response are a key element in the **requirements** for a control systems

Characteristic Parameters of the Step Response

- Steady-state value: $\bar{y} = y(\infty)$
- Settling time: $t_{s,\varepsilon}$
- Rise time: t_r
- Delay time: t_d
- Peak time: t_p
- Peak value: y_p
- Max. overshoot: $A = y_p - y(\infty)$
- Max. % overshoot:

$$\Delta\% = 100 \cdot A/y(\infty)$$
- “Period” of oscillations: T
- Damping factor: B/A



- **Case A)**

$$G(s) = \frac{\mu}{1 + s\tau}; \quad \mu > 0; \tau > 0 \quad \text{strictly proper first-order system}$$

 asymptotic stability

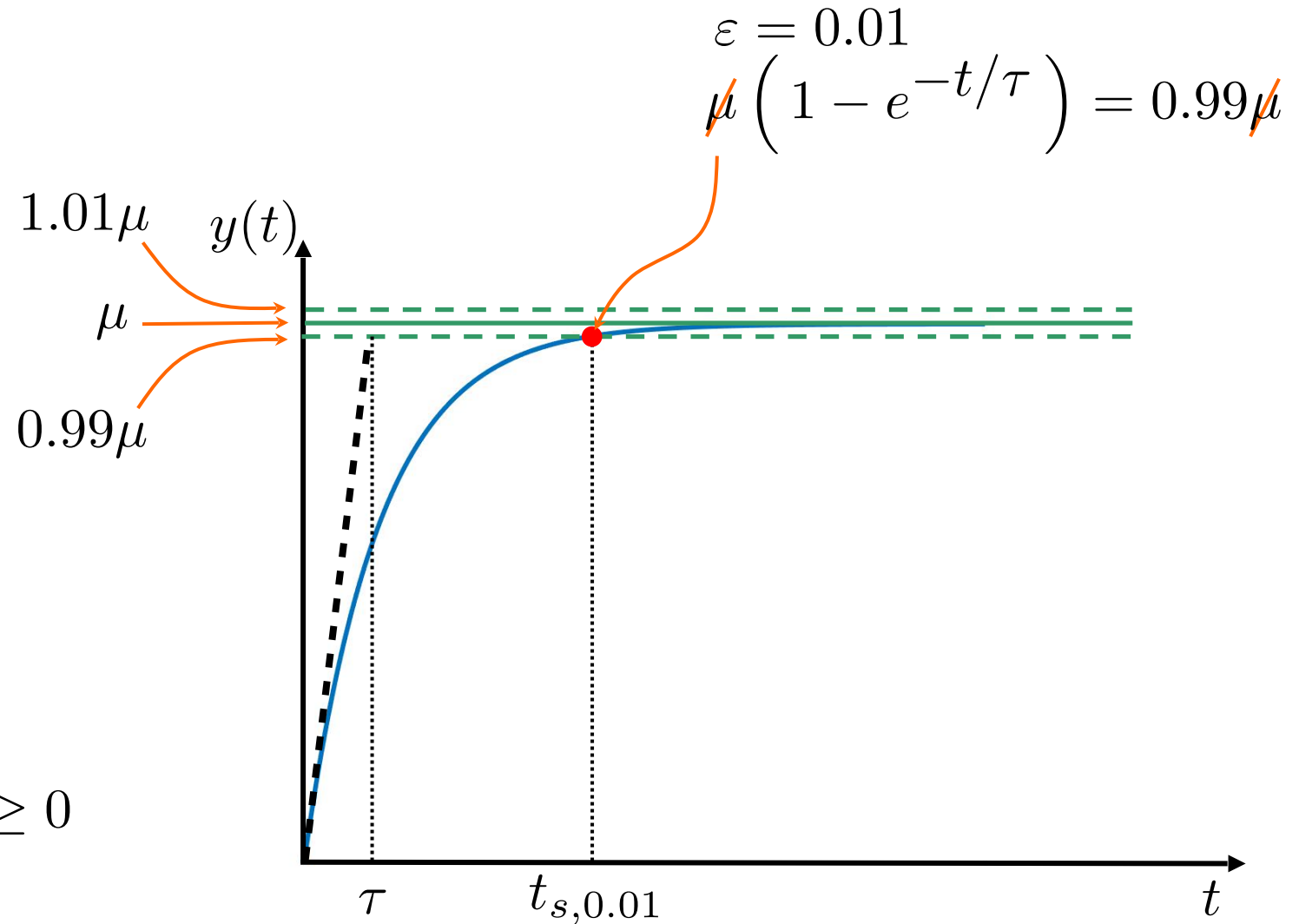
- **Case B)**

$$G(s) = \frac{\mu(1 + sT)}{1 + s\tau}; \quad \mu > 0; \tau > 0 \quad \text{non strictly proper first-order system}$$

 asymptotic stability

- **Case A)**

$$\begin{aligned}y(t) &= \mathcal{L}^{-1} \left[G(s) \frac{1}{s} \right] \\&= \mathcal{L}^{-1} \left[\frac{\mu}{s(1 + s\tau)} \right] \\&= \mathcal{L}^{-1} \left[\frac{\mu}{s} - \frac{\mu\tau}{1 + s\tau} \right] \\&= \mu \left(1 - e^{-t/\tau} \right), \quad t \geq 0\end{aligned}$$



For example, the settling time for $\varepsilon = 0.01$ can be characterised as follows:

$$1 - e^{-t/\tau} = 0.99 \quad \longrightarrow \quad e^{-t/\tau} = 0.01 \quad \longrightarrow \quad e^{t/\tau} = 100$$

 $t_{s,0.01} = \tau \ln 100 \simeq 4.6\tau$

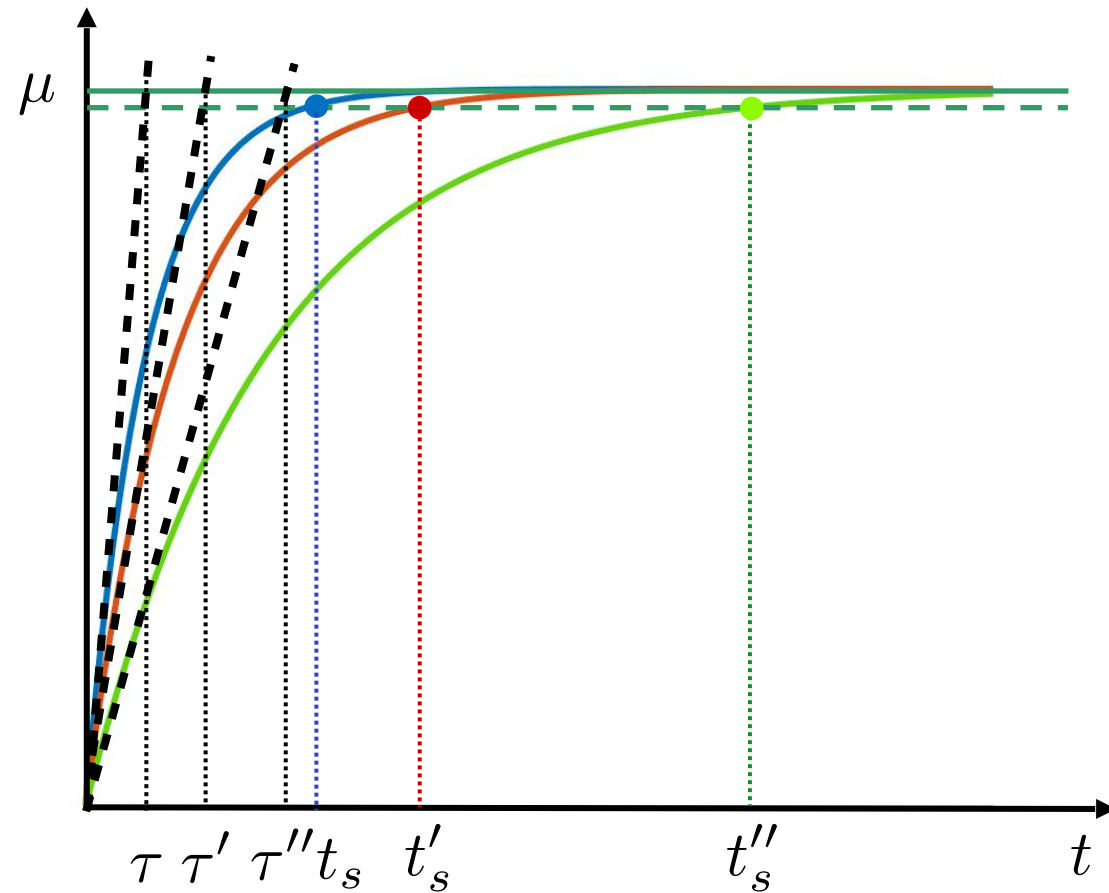
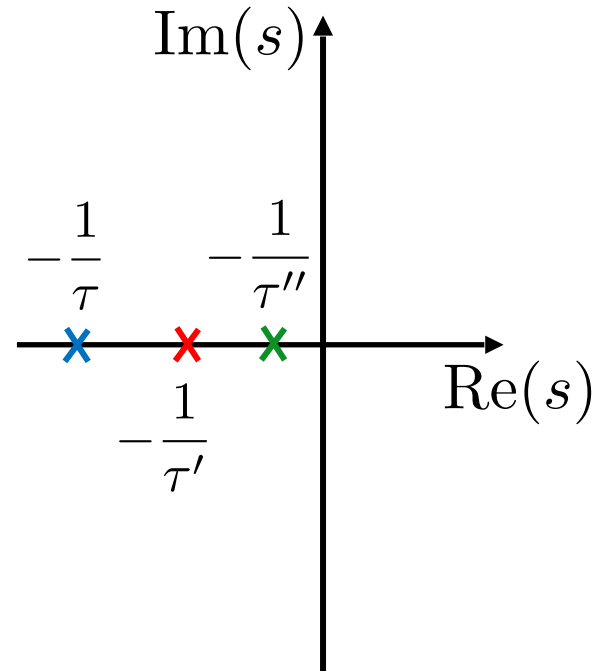
The calculation of the rising time t_r and the delay time t_d follows similar lines.

The following approximations are useful:

$t_r \simeq 2.2\tau \quad t_d \simeq 0.7\tau \quad t_{s,0.05} \simeq 3\tau \quad t_{s,0.01} \simeq 4.6\tau$

Remark: without loss of generality, from now on we shall use t_s as a shorthand for $t_{s,0.01}$

Qualitative Analysis of the Step Response

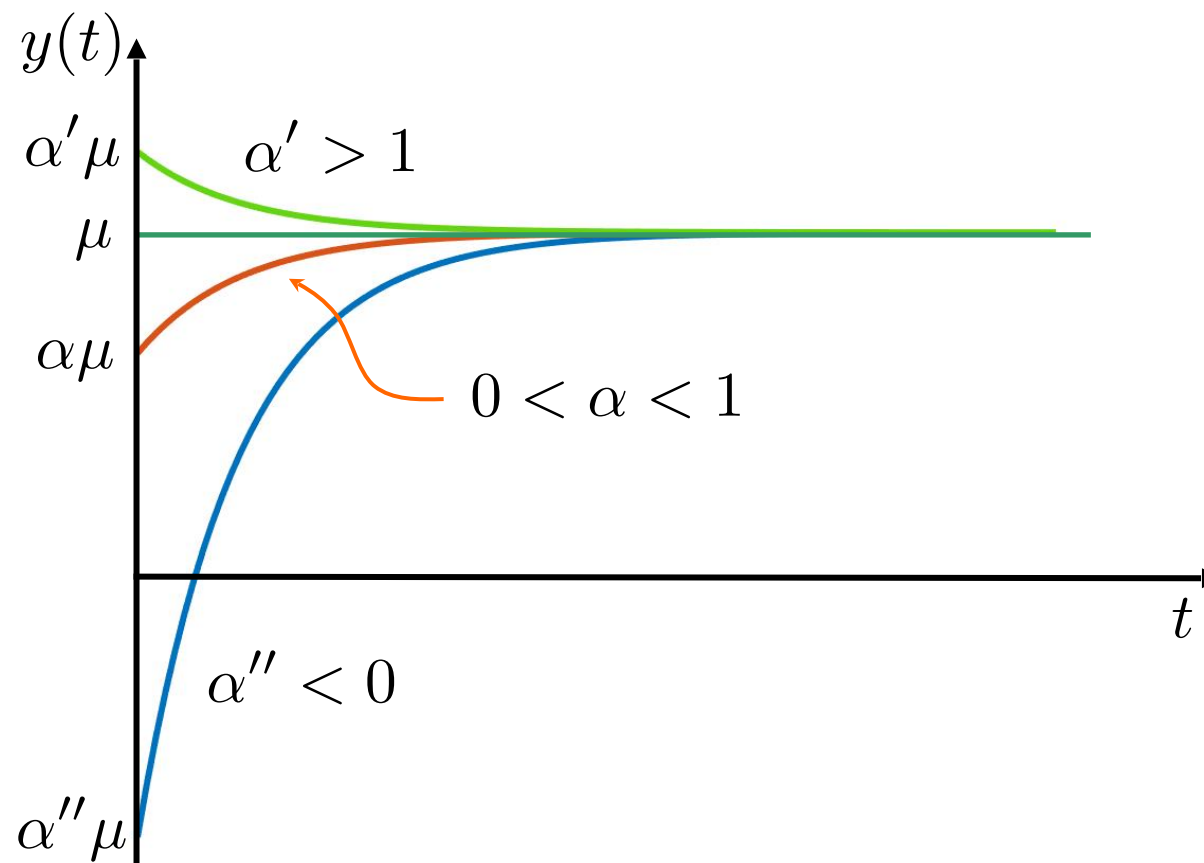
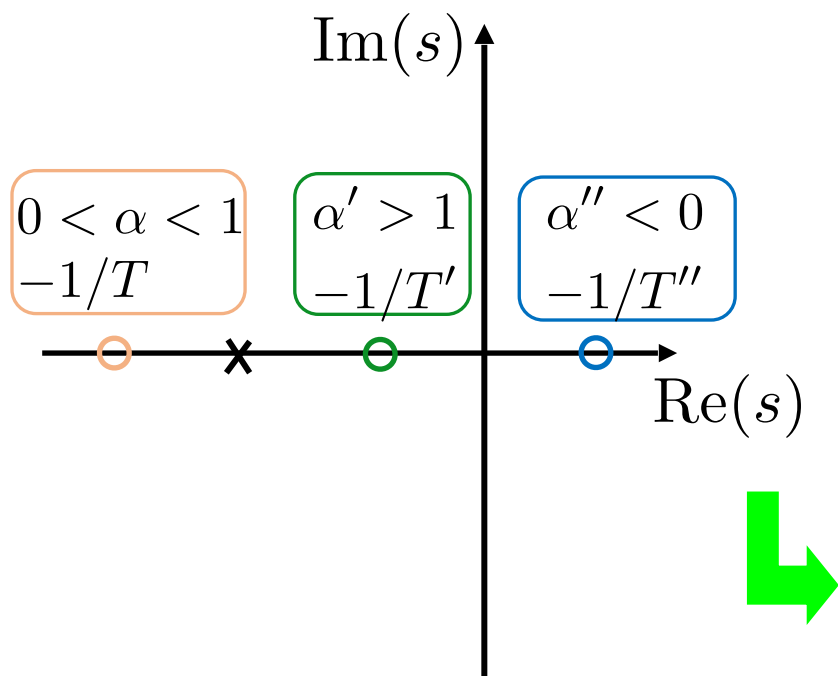


- **Case B)**

$$\begin{aligned}y(t) &= \mathcal{L}^{-1} \left[G(s) \frac{1}{s} \right] \\&= \mathcal{L}^{-1} \left[\frac{\mu(1 + sT)}{s(1 + s\tau)} \right] \\&= \mathcal{L}^{-1} \left[\frac{\mu}{s} + \frac{\mu(T - \tau)}{1 + s\tau} \right] \\&= \mu \left(1 + (\alpha - 1)e^{-t/\tau} \right), \quad t \geq 0 \text{ with } T = \alpha\tau\end{aligned}$$

Note that (the system is not strictly proper): $\lim_{t \rightarrow 0^+} y(t) = \mu \frac{T}{\tau} \neq 0$

Qualitative Analysis of the Step Response



- **Case A)**

$$G(s) = \frac{\mu}{(1 + s\tau_1)(1 + s\tau_2)} \quad \text{real poles, no zeros}$$

- **Case B)**

$$G(s) = \frac{\mu(1 + sT)}{(1 + s\tau_1)(1 + s\tau_2)} \quad \text{real poles, one zero}$$

- **Case C)**

$$G(s) = \frac{\varrho}{(s + \sigma + j\omega)(s + \sigma - j\omega)} \quad \text{complex poles, no zeros}$$

- **Case D)**

$$G(s) = \frac{\varrho(1 + sT)}{(s + \sigma + j\omega)(s + \sigma - j\omega)} \quad \text{complex poles, one zero}$$

- **Case A)**

$$G(s) = \frac{\mu}{(1 + s\tau_1)(1 + s\tau_2)} ; \quad \mu > 0 ; \quad \tau_1 \neq \tau_2$$

$$\left. \begin{array}{l} \tau_1 > 0 \\ \tau_2 > 0 \end{array} \right\} \text{asymptotic stability}$$

Without loss of generality, assume $\tau_1 > \tau_2$

$$\begin{aligned} y(t) &= \mathcal{L}^{-1} \left[G(s) \frac{1}{s} \right] = \mathcal{L}^{-1} \left[\frac{\mu}{s(1 + s\tau_1)(1 + s\tau_2)} \right] \\ &= \mathcal{L}^{-1} \left[\frac{A}{s} + \frac{B}{1 + s\tau_1} + \frac{C}{1 + s\tau_2} \right] \end{aligned}$$

where

$$\begin{aligned} A &= \left. \frac{\mu}{(1 + s\tau_1)(1 + s\tau_2)} \right|_{s=0} = \mu \\ B &= \left. \frac{\mu}{s(1 + s\tau_2)} \right|_{s=-1/\tau_1} = \frac{\mu}{-\frac{1}{\tau_1} \left(1 - \frac{\tau_2}{\tau_1}\right)} = \frac{\mu\tau_1^2}{\tau_2 - \tau_1} \\ C &= \left. \frac{\mu}{s(1 + s\tau_1)} \right|_{s=-1/\tau_2} = \frac{\mu}{-\frac{1}{\tau_2} \left(1 - \frac{\tau_1}{\tau_2}\right)} = \frac{\mu\tau_2^2}{\tau_1 - \tau_2} \end{aligned}$$

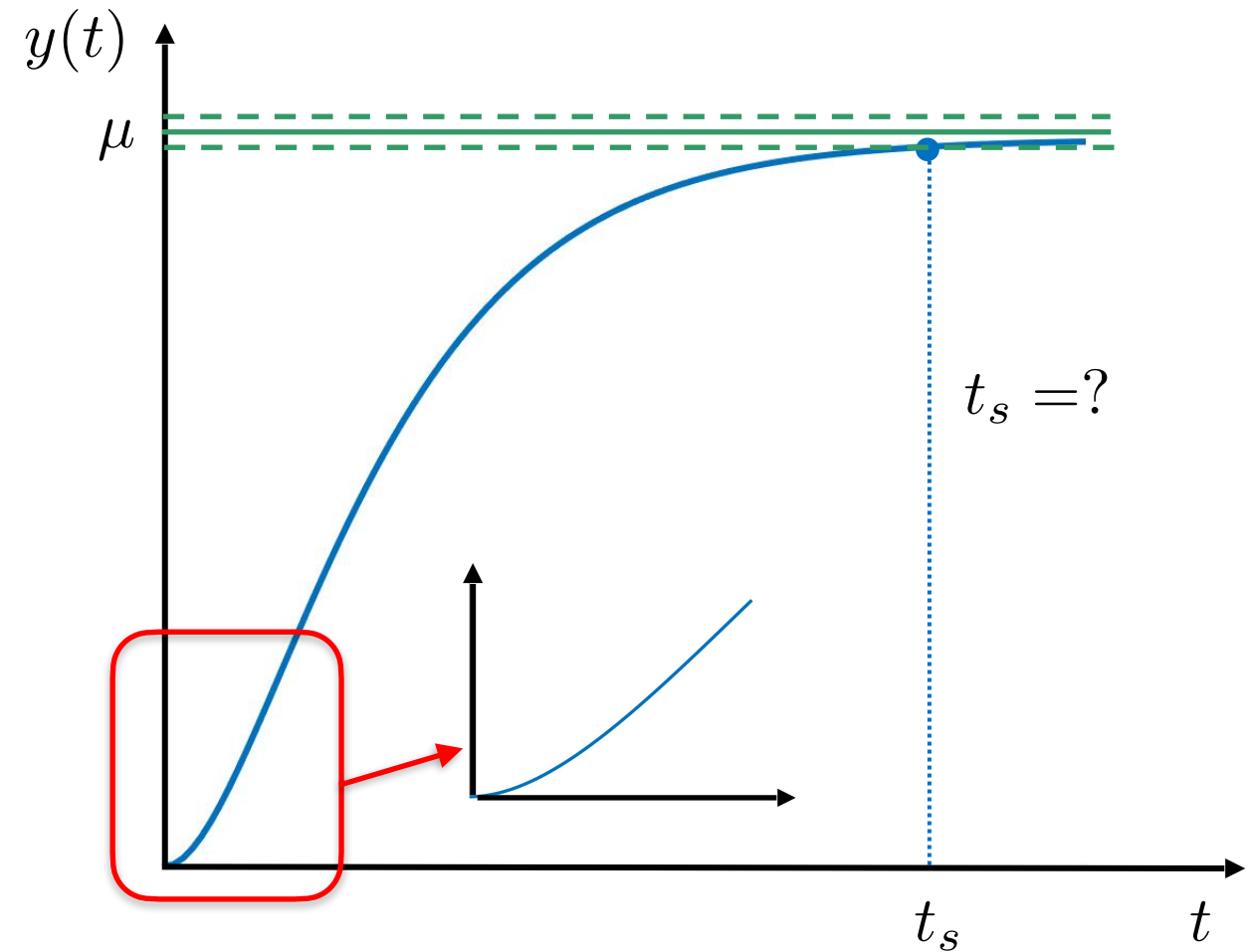
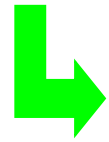
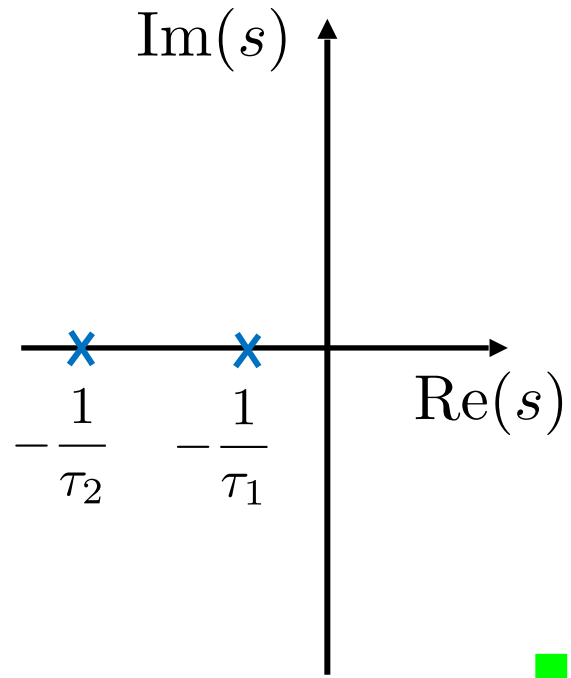
Hence:

$$\begin{aligned} y(t) &= \mathcal{L}^{-1} \left[\frac{\mu}{s} + \frac{\frac{\mu\tau_1^2}{\tau_2 - \tau_1}}{1 + s\tau_1} + \frac{\frac{\mu\tau_2^2}{\tau_1 - \tau_2}}{1 + s\tau_2} \right] \\ &= \mu \left(1 - \frac{\tau_1}{\tau_1 - \tau_2} e^{-t/\tau_1} + \frac{\tau_2}{\tau_1 - \tau_2} e^{-t/\tau_2} \right), \quad t \geq 0 \end{aligned}$$

Characteristics:

- $y(\infty) = \mu > 0$
- $y(0) = 0$
- $\dot{y}(0) = 0$
- $\ddot{y}(0) = \frac{\mu}{\tau_1\tau_2} > 0$

Qualitative Analysis of the Step Response



If $\tau_1 \gg \tau_2$:


$$y(t) = \mu \left(1 - \frac{\tau_1}{\tau_1 - \cancel{\tau_2}} e^{-t/\tau_1} + \frac{\cancel{\tau_2}}{\tau_1 - \tau_2} e^{-t/\tau_2} \right), \quad t \geq 0$$

$$\simeq \mu (1 - e^{-t/\tau_1}), \quad t \geq 0$$

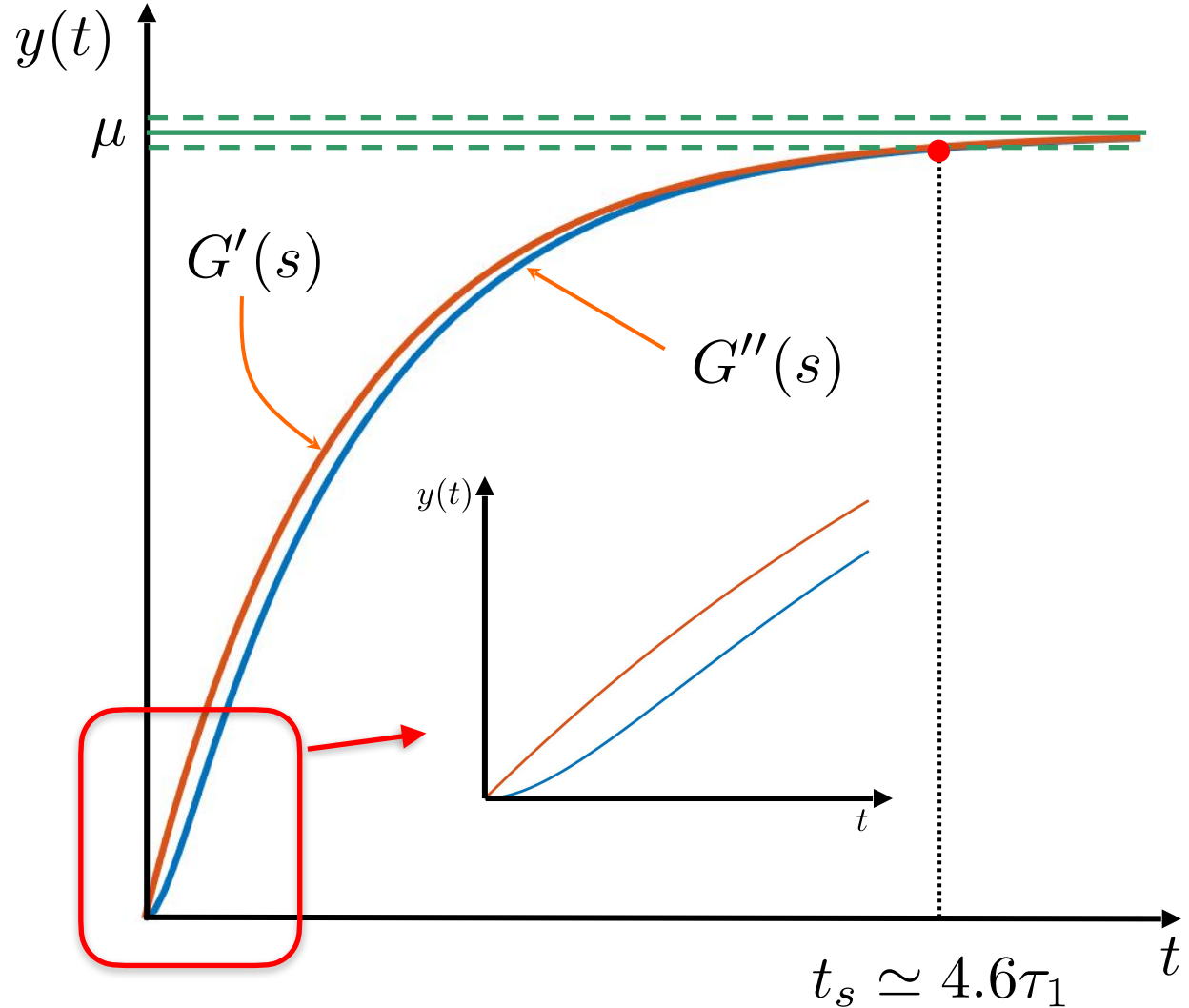

$$t_s \simeq 4.6\tau_1$$

In general, in the absence of zeros, the most influential poles on the qualitative behaviour of the step response are the ones **closer to the imaginary axis**.

$$G'(s) = \frac{\mu}{1 + s\tau_1}$$

$$G''(s) = \frac{\mu}{(1 + s\tau_1)(1 + s\tau_2)}; \tau_1 \gg \tau_2$$

- The main difference lies in the initial transient behaviour
- For a given settling time, the step-response in the second-order case without zeros has a "slower" dynamics



- **Case B)**

$$G(s) = \frac{\mu(1 + sT)}{(1 + s\tau_1)(1 + s\tau_2)} ; \quad \mu > 0 ; \quad \tau_1 \neq \tau_2$$

$$\left. \begin{array}{l} \tau_1 > 0 \\ \tau_2 > 0 \end{array} \right\} \longrightarrow \text{asymptotic stability}$$

Without loss of generality, assume $\tau_1 > \tau_2$

$$\begin{aligned} y(t) &= \mathcal{L}^{-1} \left[G(s) \frac{1}{s} \right] = \mathcal{L}^{-1} \left[\frac{\mu}{s(1 + s\tau_1)(1 + s\tau_2)} \right] \\ &= \mathcal{L}^{-1} \left[\frac{A}{s} + \frac{B}{1 + s\tau_1} + \frac{C}{1 + s\tau_2} \right] \end{aligned}$$

where

$$\begin{aligned} A &= \left. \frac{\mu(1 + sT)}{(1 + s\tau_1)(1 + s\tau_2)} \right|_{s=0} = \mu \\ B &= \left. \frac{\mu(1 + sT)}{s(1 + s\tau_2)} \right|_{s=-1/\tau_1} = \frac{\mu(1 - T/\tau_1)}{-\frac{1}{\tau_1}(1 - \frac{\tau_2}{\tau_1})} = \frac{\mu\tau_1(\tau_1 - T)}{\tau_2 - \tau_1} \\ C &= \left. \frac{\mu(1 + sT)}{s(1 + s\tau_1)} \right|_{s=-1/\tau_2} = \frac{\mu(1 - T/\tau_2)}{-\frac{1}{\tau_2}(1 - \frac{\tau_1}{\tau_2})} = \frac{\mu\tau_2(\tau_2 - T)}{\tau_1 - \tau_2} \end{aligned}$$

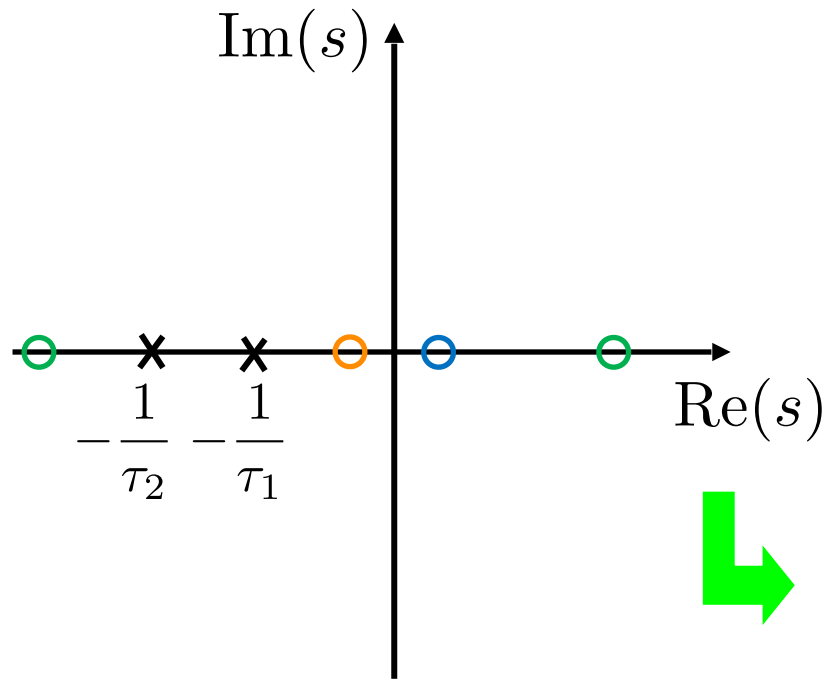
Hence:

$$y(t) = \mu \left(1 - \frac{\tau_1 - T}{\tau_1 - \tau_2} e^{-t/\tau_1} + \frac{\tau_2 - T}{\tau_1 - \tau_2} e^{-t/\tau_2} \right), \quad t \geq 0$$

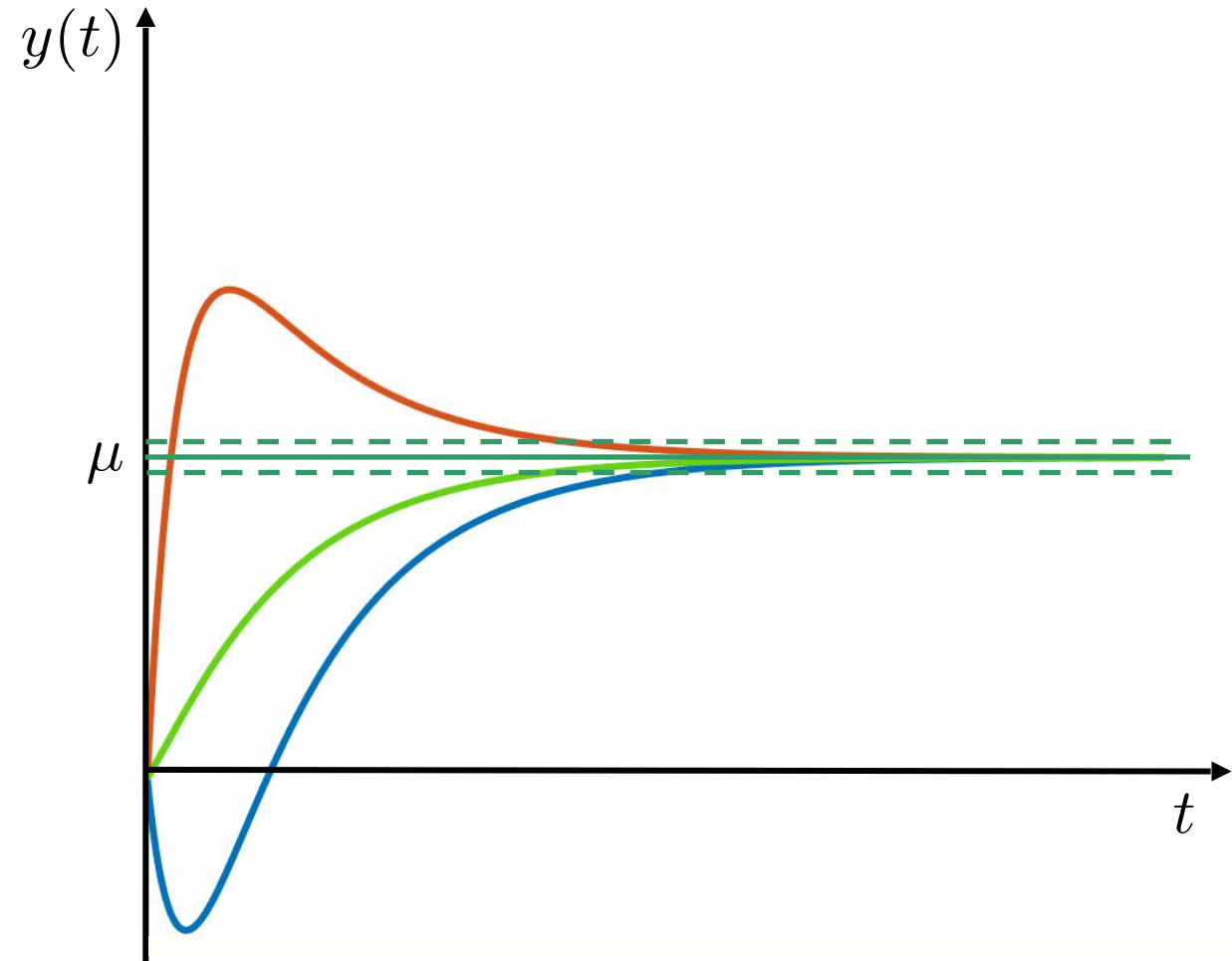
Characteristics:

- $y(\infty) = \mu > 0$
- $y(0) = 0$
- $\dot{y}(0) = \frac{\mu T}{\tau_1 \tau_2} \begin{cases} > 0, & \text{if } T > 0 \\ < 0, & \text{if } T < 0 \end{cases}$

Qualitative Analysis of the Step Response



- zero with little influence
- overshoot
- undershoot



- **Case C)**

$$G(s) = \frac{\varrho}{(s + \sigma + j\omega)(s + \sigma - j\omega)}$$

$$\mu = G(0) = \frac{\varrho}{\sigma^2 + \omega^2}$$

poles: $-\sigma \pm j\omega$

$\sigma > 0$  asymptotic stability

$$\omega > 0$$

$$\varrho > 0$$

$$Y(s) = \frac{G(s)}{s} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\sigma s + \sigma^2 + \omega^2}$$


$$As^2 + 2A\sigma s + A\sigma^2 + A\omega^2 + Bs^2 + Cs = \varrho$$


$$\begin{cases} A + B = 0 \\ 2A\sigma + C = 0 \\ A(\sigma^2 + \omega^2) = \varrho \end{cases} \quad \longrightarrow \quad \begin{cases} A = \frac{\varrho}{\sigma^2 + \omega^2} = \mu \\ B = -\mu \\ C = -2\sigma\mu \end{cases}$$


$$\begin{aligned} Y(s) &= \mu \left[\frac{1}{s} - \frac{s + 2\sigma}{s^2 + 2\sigma s + \sigma^2 + \omega^2} \right] = \mu \left[\frac{1}{s} - \frac{s + \sigma + \sigma}{(s + \sigma)^2 + \omega^2} \right] \\ &= \mu \left[\frac{1}{s} - \frac{s + \sigma}{(s + \sigma)^2 + \omega^2} - \frac{\sigma}{\omega} \frac{\omega}{(s + \sigma)^2 + \omega^2} \right] \end{aligned}$$

$$\text{Hence: } y(t) = \mu \left[1 - e^{-\sigma t} \cos(\omega t) - \frac{\sigma}{\omega} e^{-\sigma t} \sin(\omega t) \right], \quad t \geq 0$$

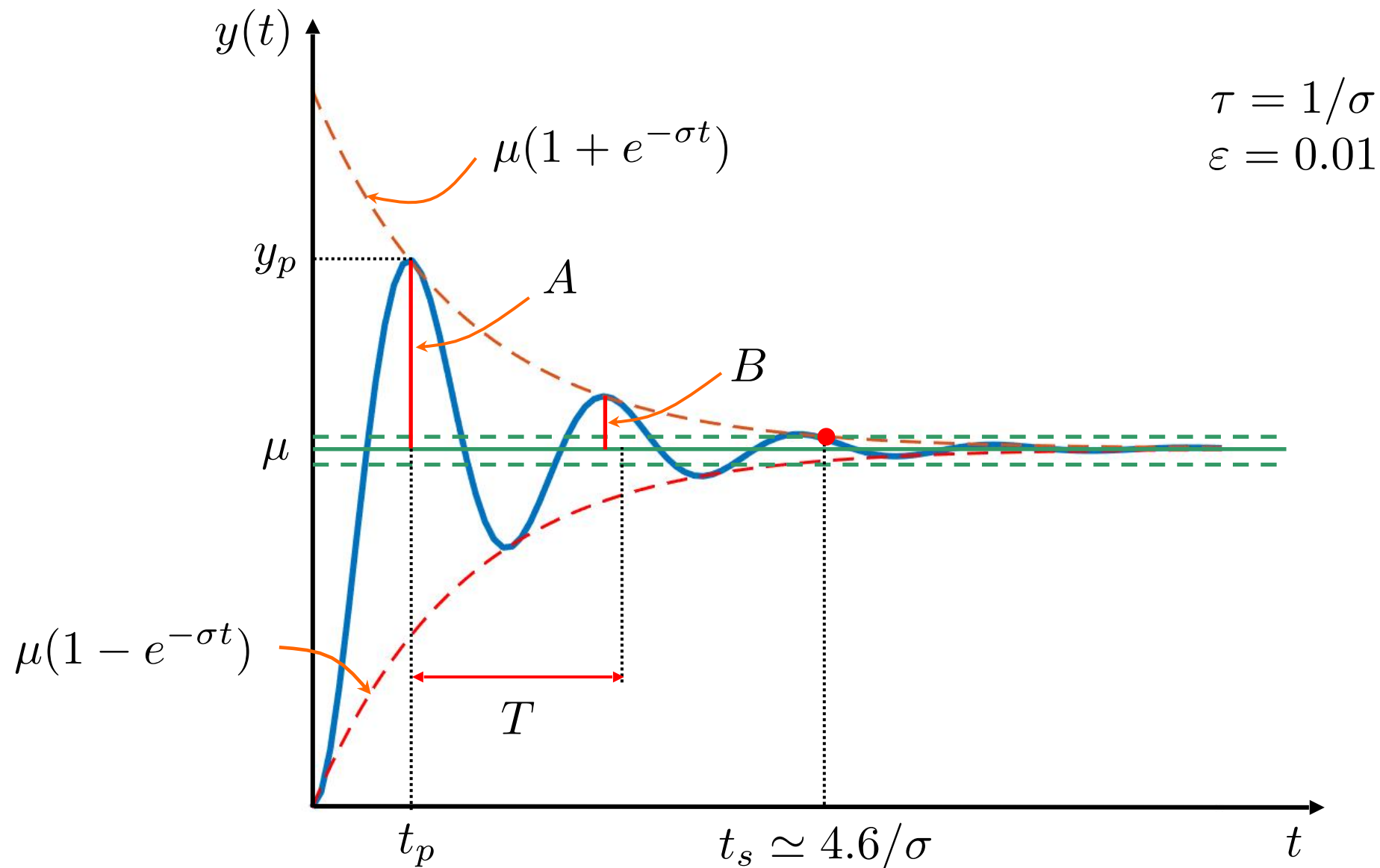
$$= \mu \left[1 - e^{-\sigma t} \left(\cos(\omega t) + \frac{\sigma}{\omega} \sin(\omega t) \right) \right], \quad t \geq 0$$

$$= \mu \left[1 - \underbrace{\frac{\sqrt{\sigma^2 + \omega^2}}{\omega} e^{-\sigma t} \sin(\omega t + \varphi)}_{\text{damped oscillations}} \right], \quad t \geq 0$$

$$\text{where } \varphi = \arccos \left(\frac{\sigma}{\sqrt{\sigma^2 + \omega^2}} \right)$$

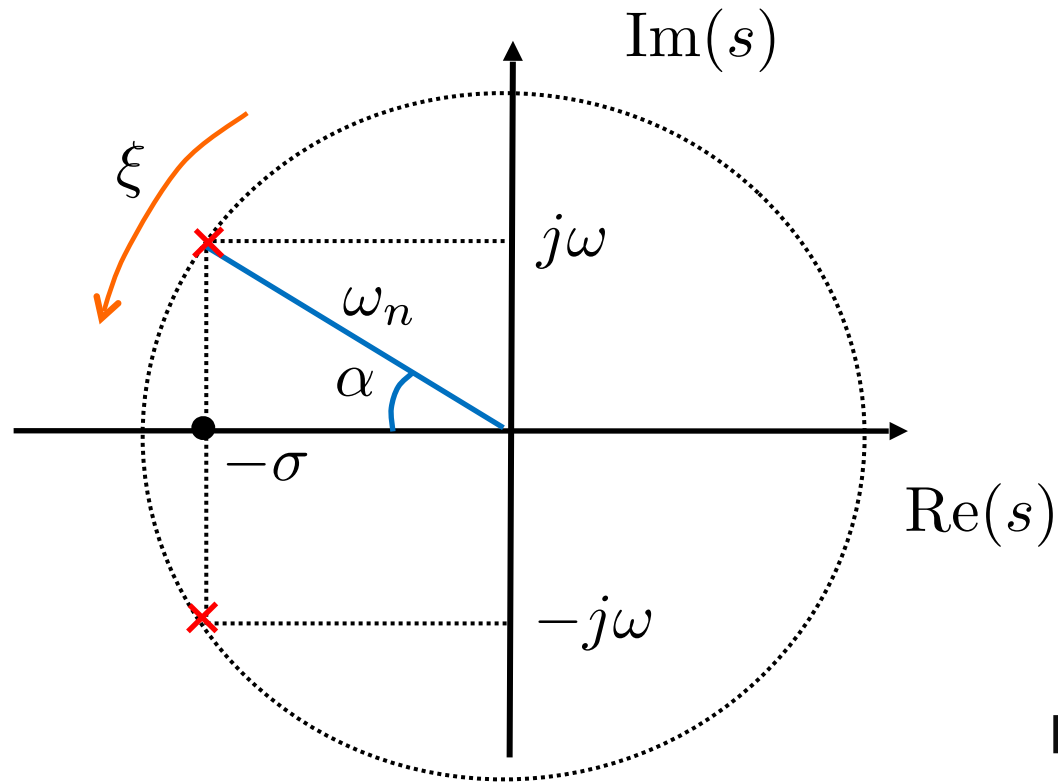
- Characteristics:
- $y(\infty) = \mu > 0$
 - $y(0) = 0$
 - $\dot{y}(0) = 0$
 - $\ddot{y}(0) = \varrho > 0$

Qualitative Analysis of the Step Response



Characteristic Parameters of the Step Response

Recall from Part 4:



$$G(s) = \frac{\varrho}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\omega_n^2 = \sigma^2 + \omega^2$$

$$\omega_n \xi = \sigma$$

$$\omega_n \sqrt{1 - \xi^2} = \omega$$

Parameters:

ω_n natural angular frequency:

$\xi = \cos(\alpha)$ damping ratio

and:

$$\begin{aligned} G(s) &= \frac{\varrho}{(s + \sigma + j\omega)(s + \sigma - j\omega)} = \frac{\varrho}{(s + \sigma)^2 + \omega^2} \\ &= \frac{\varrho}{s^2 + \underbrace{2\sigma s}_{2\xi\omega_n} + \underbrace{\sigma^2 + \omega^2}_{\omega_n^2}} = \frac{\varrho}{s^2 + 2\xi\omega_n s + \omega_n^2} \end{aligned}$$


$$G(s) = \frac{\varrho/\omega_n^2}{1 + \frac{2\xi}{\omega_n}s + \frac{1}{\omega_n^2}s^2} = \frac{\mu}{1 + \frac{2\xi}{\omega_n}s + \frac{1}{\omega_n^2}s^2}$$

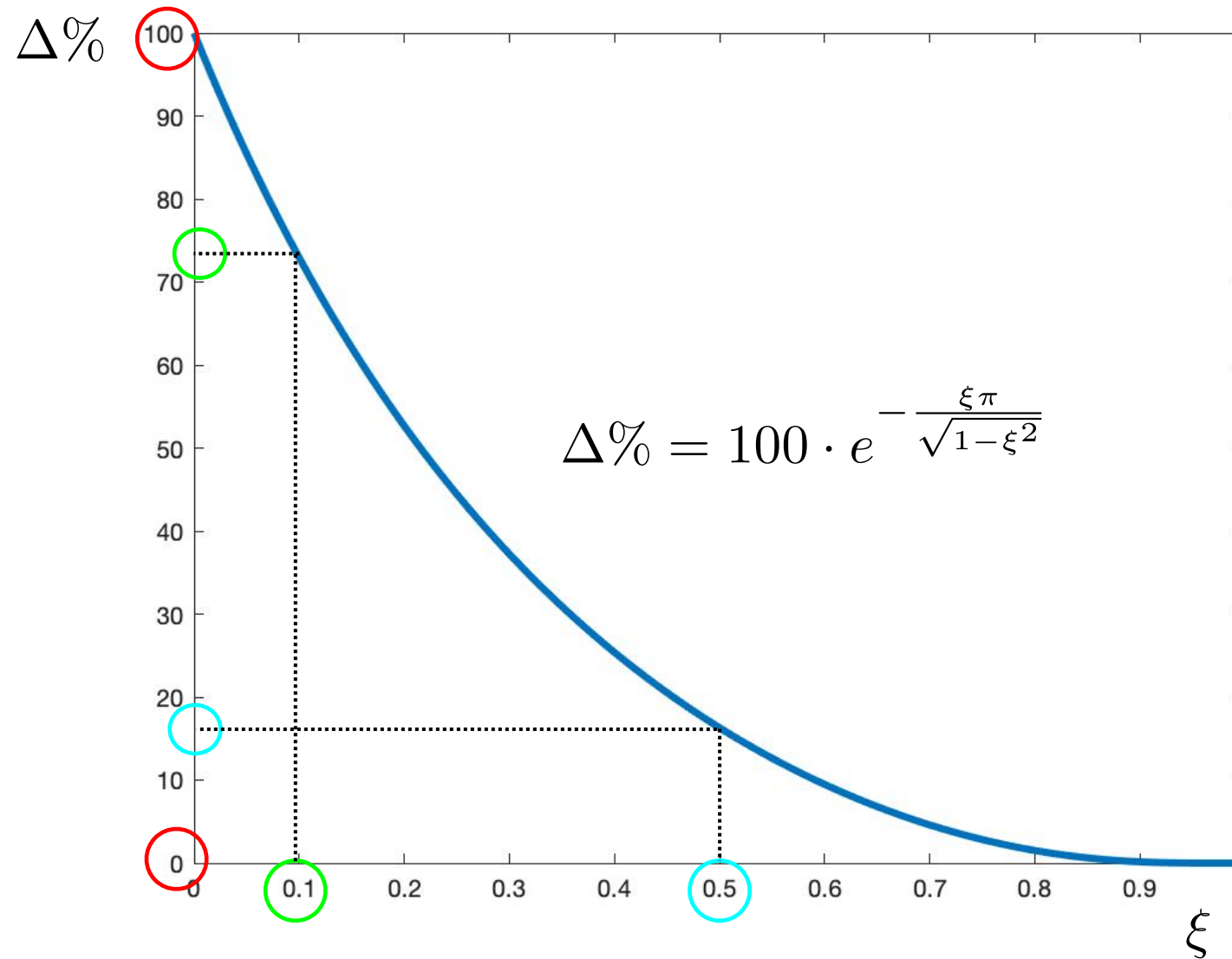
where: $\mu := \frac{\varrho}{\omega_n^2}$

Hence:

- Settling time: $t_s \simeq \frac{4.6}{\sigma} = \frac{4.6}{\xi\omega_n}$
- Peak time: $t_p = \frac{\pi}{\omega} = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$
- Peak value: $y_p = \mu \left[1 + e^{-\frac{\sigma\pi}{\omega}} \right] = \mu \left[1 + e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \right]$
- Maximum percentage overshoot: $\Delta\% = 100 \cdot \frac{A}{\mu} = e^{-\sigma\pi/\omega} = 100 \cdot e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$
- “Period” of oscillations: $T = \frac{2\pi}{\omega} = \frac{2\pi}{\omega_n \sqrt{1-\xi^2}}$
- Damping factor: $\frac{B}{A} = \dots = \Delta^2 = e^{-2\sigma\pi/\omega} = e^{-\frac{2\xi\pi}{\sqrt{1-\xi^2}}}$

only depend on ξ
but **not** on ω_n

Maximum Percentage Overshoot



- **No damping:** $\xi = 0$

$$G(s) = \frac{\varrho}{s^2 + \omega_n^2} \quad \text{poles: } \pm j\omega_n$$



Undamped oscillations

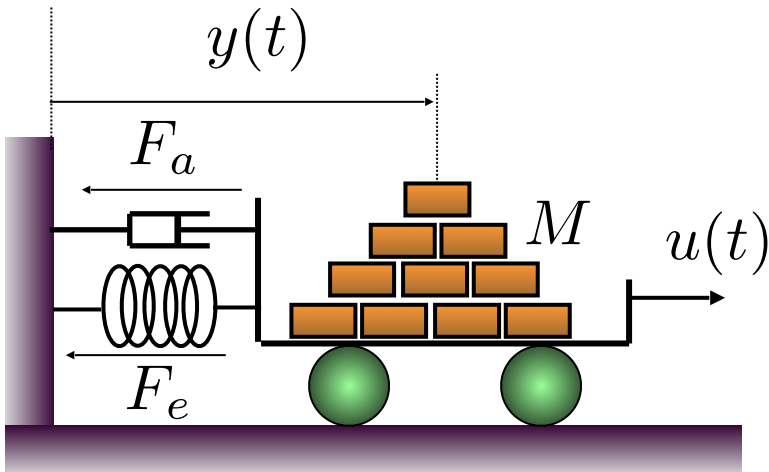
- **Full damping:** $\xi = 1$

$$G(s) = \frac{\varrho}{(s + \omega_n)^2} \quad \text{poles: } -\omega_n; -\omega_n$$



No oscillations at all

Example 1



Hence:

$$\begin{aligned}\mu &= G(0) = \frac{1}{k} \\ 2\xi\omega_n &= \frac{h}{M} \\ \omega_n^2 &= \frac{k}{M}\end{aligned}$$

$$\begin{aligned}G(s) &= \frac{1}{Ms^2 + hs + k} \\ &= \frac{1/M}{s^2 + \frac{h}{M}s + \frac{k}{M}}\end{aligned}$$

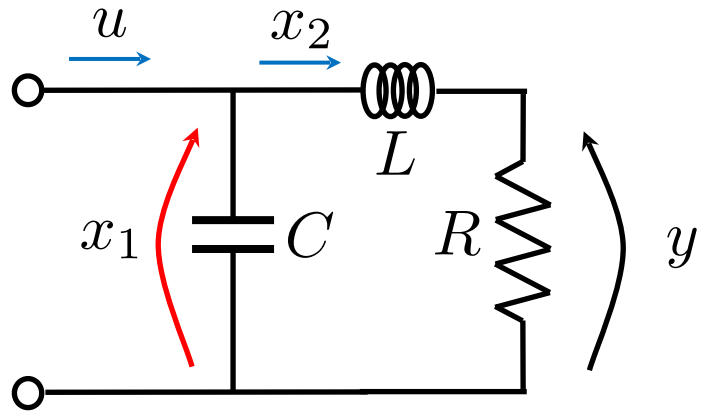


$$s^2 + 2\xi\omega_n s + \omega_n^2 = \frac{1/M}{s^2 + \frac{h}{M}s + \frac{k}{M}}$$



$$\begin{aligned}\omega_n &= \sqrt{\frac{k}{M}} \\ \xi &= \frac{h}{2\sqrt{kM}}\end{aligned}$$

Example 2



$$\begin{cases} C\dot{x}_1 = u - x_2 \\ L\dot{x}_2 = x_1 - Rx_2 \\ y = Rx_2 \end{cases}$$

$$\rightarrow A = \begin{bmatrix} 0 & -1/C \\ 1/L & -R/L \end{bmatrix} \quad B = \begin{bmatrix} 1/C \\ 0 \end{bmatrix} \quad C = [0 \quad R]$$

$$\rightarrow G(s) = [0 \quad R] \begin{bmatrix} s & 1/C \\ -1/L & s + R/L \end{bmatrix}^{-1} \begin{bmatrix} 1/C \\ 0 \end{bmatrix} = \dots = \frac{R/(LC)}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$\rightarrow \omega_n = \frac{1}{\sqrt{LC}}; \quad \xi = \frac{R}{2} \sqrt{\frac{C}{L}}; \quad \mu = R$$

- **Case D)**

$$G(s) = \frac{\mu(1 + sT)}{1 + \frac{2\xi}{\omega_n}s + \frac{1}{\omega_n^2}s^2}; \quad 0 < \xi < 1; \quad \omega_n > 0; \quad \mu > 0$$

Characteristics of the step response:

- $y(\infty) = \mu > 0$
- $y(0) = 0$
- $\dot{y}(0) = \mu T \omega_n^2 \begin{cases} > 0, & \text{if } T > 0 \\ < 0, & \text{if } T < 0 \end{cases}$

Qualitative Analysis:

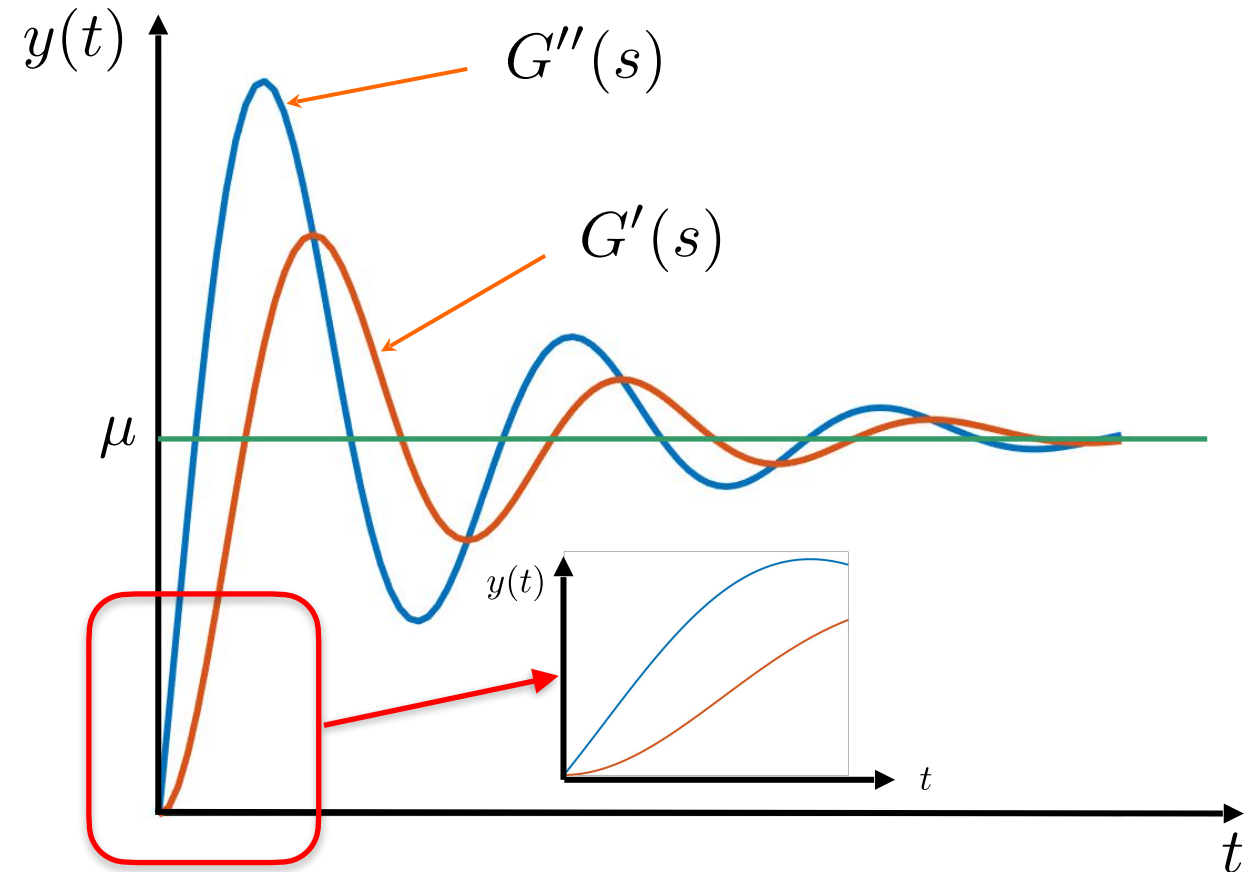
Comparison between Case C) (no zeros) and Case D) (one zero)



$$G'(s) = \frac{\mu}{1 + \frac{2\xi}{\omega_n}s + \frac{1}{\omega_n^2}s^2}$$

$$G''(s) = \frac{\mu(1 + sT)}{1 + \frac{2\xi}{\omega_n}s + \frac{1}{\omega_n^2}s^2}$$

- Again, the main difference lies in the initial transient behaviour
- For a given settling time, the step-response in Case C) without zeros has a "slower" dynamics



For simplicity, consider the case of real poles only:

$$G(s) = \frac{\mu}{s^g} \frac{\prod_{i=1}^m (1 + sT_i)}{\prod_{i=1}^n (1 + s\tau_i)}$$

Recall (in the absence of common factors in $G(s)$):

Asymptotic Stability $\longleftrightarrow$ $\begin{matrix} \text{Re}(\text{poles}) < 0 \\ g \leq 0 \end{matrix}$

$$y(t) = \mathcal{L}^{-1} \left[G(s) \frac{1}{s} \right]$$

- **Initial Value Theorem**

$$\lim_{t \rightarrow 0^+} y(t) = \lim_{s \rightarrow \infty} \cancel{s} \frac{1}{\cancel{s}} G(s) \begin{cases} = 0, & \text{if } m < n \\ \neq 0, & \text{if } m = n \end{cases}$$

- **Final Value Theorem**

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} \cancel{s} \frac{1}{\cancel{s}} G(s) \begin{cases} = \mu, & \text{if } g = 0 \\ = 0, & \text{if } g < 0 \end{cases}$$

Again, for simplicity, consider the case of real poles:

$$Y(s) = G(s) \frac{1}{s} = \frac{\alpha_0}{s} + \frac{\alpha_1}{1 + s\tau_1} + \dots + \frac{\alpha_n}{1 + s\tau_n}$$


$$y(t) = \mathcal{L}^{-1} \left[G(s) \frac{1}{s} \right]$$

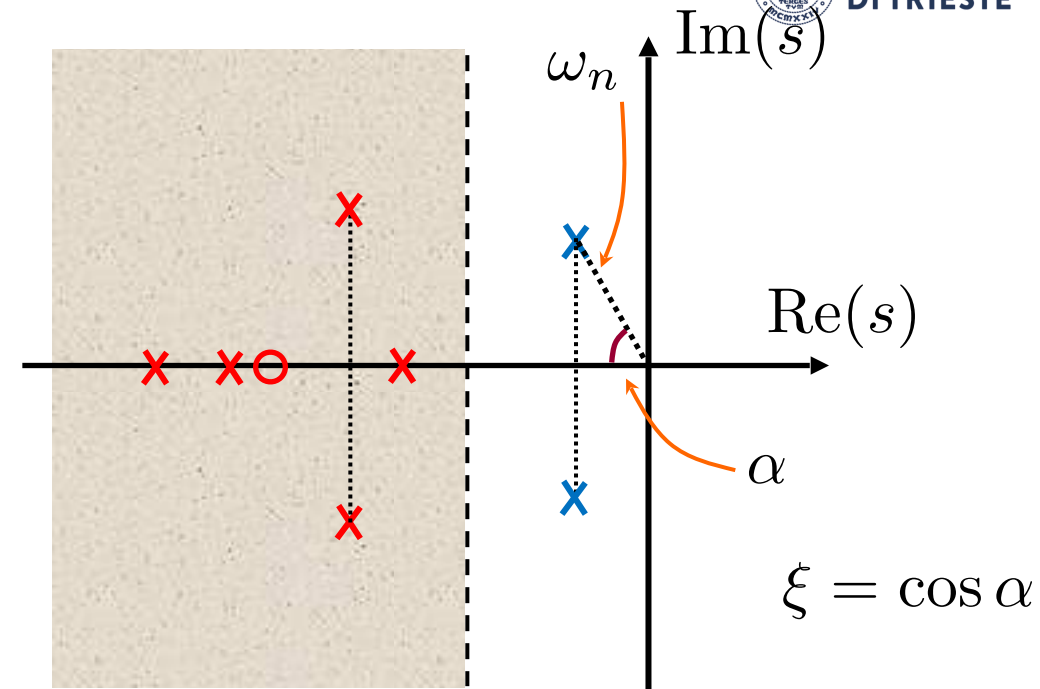
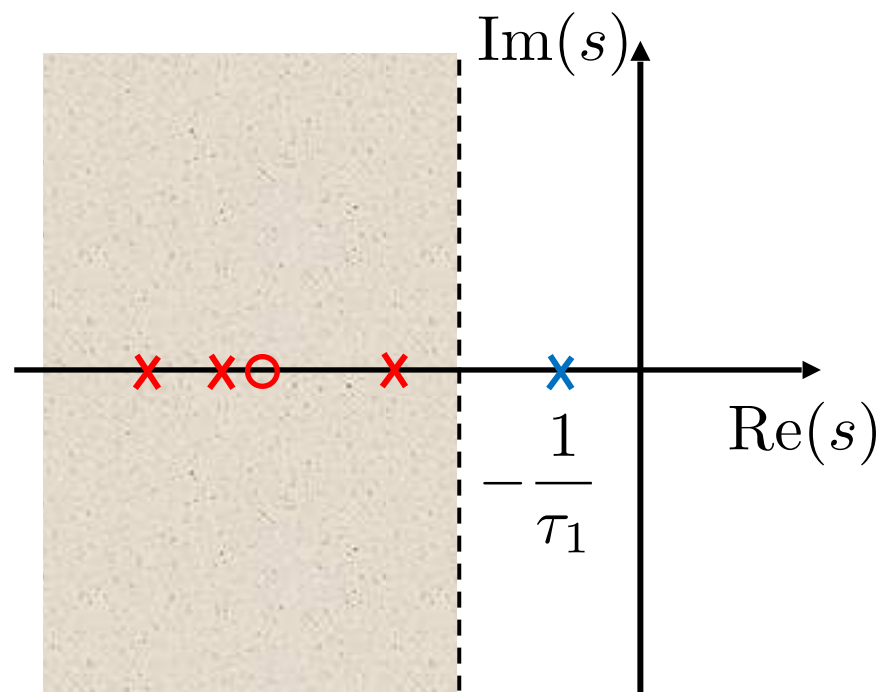
$$= \alpha_0 + \frac{\alpha_1}{\tau_1} e^{-t/\tau_1} + \dots + \frac{\alpha_n}{\tau_n} e^{-t/\tau_n}$$

Assuming: $\tau_1 > \tau_2 > \dots > \tau_n$


$$y(t) = \alpha_0 + \frac{\alpha_1}{\tau_1} e^{-t/\tau_1} + \dots + \frac{\alpha_n}{\tau_n} e^{-t/\tau_n}$$

$$\simeq \alpha_0 + \frac{\alpha_1}{\tau_1} e^{-t/\tau_1} \quad \text{dominant component, hence: } t_s \simeq 4.6\tau_1$$

Dominant Poles Approximation: Real Poles



- When using the **dominant poles** approximation:
 - It is important to “preserve” the gain
 - Zeros located close to the imaginary axis have to be properly taken into account
- This approximation is useful in **qualitative** analysis and the for initial and rough controller's design steps

Example

$$G(s) = \frac{400(1 + s)}{(1 + 0.2s)(1 + 0.1s)(s^2 + 2s + 4)}$$

↓ $\omega_n = 2$
 $\xi = 1/2$

$$\mu = G(0) = 100$$

poles: -5
 -10
 $-1 \pm j\sqrt{3}$

zero: -1

